

RESEARCH ARTICLE

Deciphering Key Descriptors for Scaling Relationships in Graphene-Supported Pt_n Clusters via Machine Learning

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ABSTRACT

Subnano clusters have emerged as a promising class of electrocatalysts, enabling efficient utilization of noble metals and superior activity for key electrochemical reactions. However, their fluxional nature and complex structure–activity relationships give rise to non-monotonic catalytic behavior, making activity evaluation highly challenging. Here, we develop a machine learning (ML) framework to assess the size-dependent activity of graphene-supported Pt_n ($n = 7 - 13$) subnano clusters for the oxygen reduction reaction (ORR). Using non-ab initio descriptors, the trained ML model accurately predicts site-specific adsorption energies of key ORR intermediates across heterogeneous active sites. Feature-driven uncertainty quantification highlights the irregular catalytic activity and the importance of ensemble representation, while identifying the key geometric descriptors responsible for breaking the scaling relationship in the subnanometer regime. Furthermore, we uncover critical insights into the dynamics of active electrocatalysts under ambient ORR conditions on supports, showing their strong oxidative tendencies that drive monolayer formation of intermediates. We also derive reaction networks for the rate-determining steps, pinpointing the saturation onset point that signals changes in binding sites. Unlike bare clusters, this study provides an atomistic understanding of fluxional dynamics and structural evolution on supports, offering guidelines for the rational design of highly efficient electrocatalysts for fuel cell applications.

1 | Introduction

Electrochemical energy conversion technologies are widely regarded as one of the most promising solutions to address global energy demands and environmental challenges within a sustainable, carbon-neutral framework [1]. Among them, fuel cells [2] and metal-air batteries [3] stand out as potential candidates for clean energy applications. However, their large-scale deployment is hindered by the sluggish kinetics of the oxygen reduction reaction (ORR) at the cathode, which leads to high overpotentials and reduced efficiency. To this end, platinum-based electrocatalysts are valued for their high activity for ORR;

however, their widespread commercialization is limited by high cost and vulnerability to poisoning during long-term operation, motivating the search for low-cost alternatives [4].

Subnanometer clusters have recently drawn significant attention as a distinct class of electrocatalysts because of their high atomic utilization and remarkable catalytic properties. Unlike bulk materials, these clusters at finite temperatures are characterized by a relatively flat potential energy surface, which imparts fluxionality and non-Arrhenius behavior [5]. Their large fraction of under-coordinated sites contributes to dynamic structural changes, resulting in size-dependent, non-monotonic catalytic activity [6].

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In this non-scalable regime, cluster structures and electronic distributions are highly complex, giving rise to metastability-driven reactivity. Therefore, accurate modelling of their catalytic performance requires ensemble representations that capture the diversity of accessible states [7]. Importantly, the interaction between subnano clusters and their supports can locally reshape coordination environments, modify conventional scaling relationships, and create opportunities to overcome the volcano-plot limitations [8]. Despite these promises, systematic evaluation of how different sizes of supported subnano clusters impact the catalytic activity remains limited. This gap reflects both the intrinsic complexity of the non-scalable regime and the challenges of performing high-throughput ab initio simulations in a size range where “every atom counts”.

Machine learning (ML) has emerged as a powerful tool for accelerating catalyst discovery by capturing complex interactions between catalysts and adsorbates, while unravelling the multi-dimensional structure-property relationships [9]. ML algorithms have been applied to a wide range of electrochemical reactions, enabling faster discovery and optimization of catalysts across vast chemical spaces [10, 11]. Descriptor-driven frameworks have also been used to identify cost-effective Pt-free catalysts, including single-atom and dual-atom systems as well as high-entropy alloys [12, 13]. While these advances demonstrate the promise of ML in catalyst discovery, their applications to subnanometer regime clusters remain limited. In particular, the factors governing adsorption energies and the catalytic behavior of supported subnano clusters are not fully understood. Additionally, systematic studies of the multifidelity features contributing to the non-scaling relationships arising from intermediate adsorption at undercoordinated active sites are still in their infancy and require in-depth data-driven explorations.

Herein, we present a combined DFT-ML study on the ORR activity of size-selected Pt_n clusters ($n = 7 - 13$) supported on graphene (Pt_n/G), as shown in Scheme 1.

To model the electrodes with graphene-supported subnano clusters Pt_n/G ($n = 7 - 13$), we performed systematic global optimizations to sample the potential energy surface and extract an ensemble of low-energy isomers for catalytic investigations. We demonstrate that the fluxional nature of these clusters leads to strong variations in intermediate interactions, resulting in a complex, non-monotonic structure-activity relationship. Using the Sabatier principle as an optimal screening criterion, we trained ML models to predict ORR activity from non-ab initio features that capture the interplay of geometric and electronic environments. A feature-driven interpretability analysis reveals that the geometric descriptors of heterogeneous active sites play a crucial role in breaking scaling relationships, a finding further validated by the uncertainty principle. The ML predictions were validated, active electrocatalysts were screened, and the ORR mechanism was derived following the adsorbate evolution. We further examined low-coverage-based active catalysts to construct reaction networks and phase diagrams at higher intermediate coverages. Overall, our combined DFT and ML explorations provide insight into the non-scaling behavior, fluxional dynamics, and coverage-dependent ORR activity of graphene-supported subnano clusters through combined DFT and ML explorations. The graphene support used

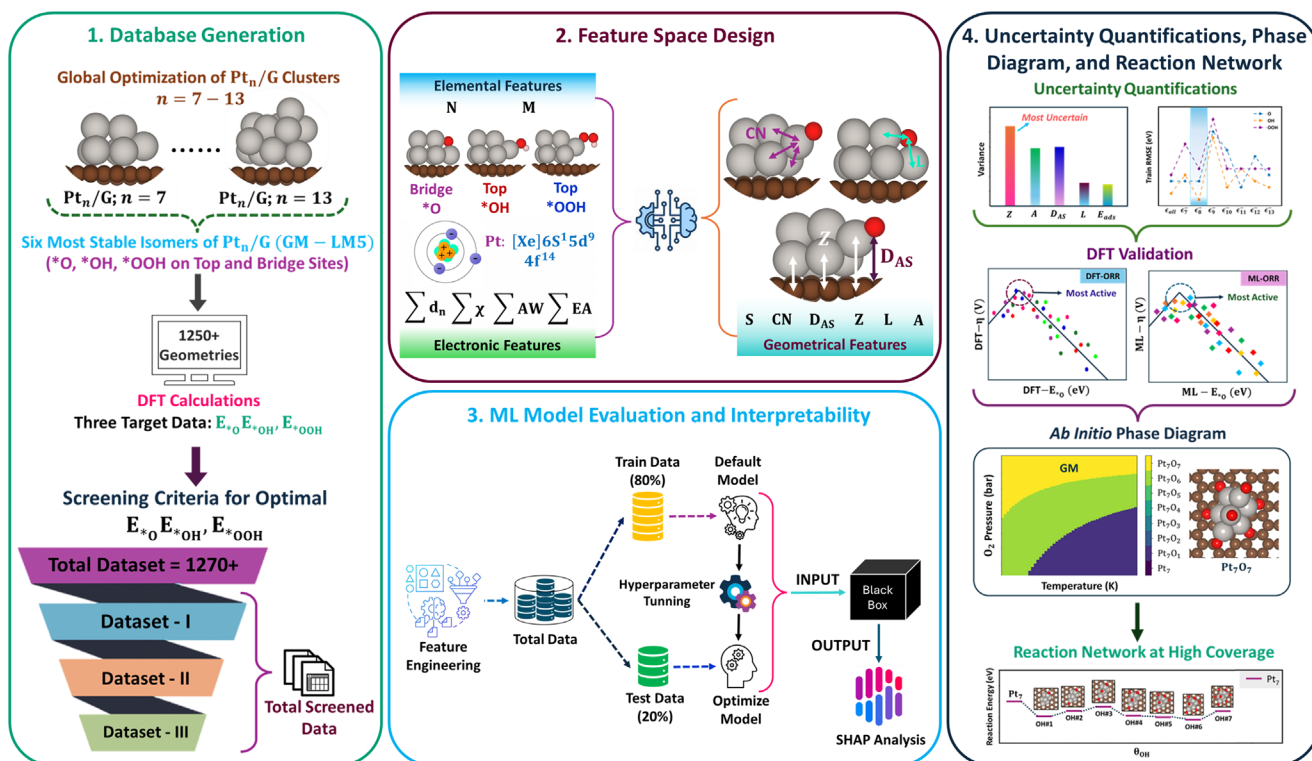
in our investigation for anchoring Pt clusters has no carbon vacancies.

2 | Methods

2.1 | Structural Design of Pt_n/G Subnano Electrocatalysts

To investigate the size-dependent ORR activity of Pt_n/G ($n = 7 - 13$) subnano clusters, we first extracted representative structures by exhaustively sampling the energy landscape of each size. Global optimization searches were performed for Pt clusters containing 7–13 atoms on graphene using the Parallel Global Optimization Pathway Toolkit to identify the global minima (GM) and low-energy local minima (LM) configurations [14]. Multiple states for each size were generated using the bond-length distribution algorithm approach, and unique geometries were extracted through the depth-first search method. For their local optimization, plane-wave spin-polarized DFT calculations were employed in the Vienna ab initio simulation package (VASP) [15] with the Perdew–Burke–Ernzerhof (PBE) functional within the generalized gradient approximation (GGA) [16]. Full computational details are provided in Text S1. The thermodynamic energy distributions of the sampled states, referenced to the GM energy ($\Omega = E_{\text{LM}} - E_{\text{GM}}$), are shown in Figure 1a, with the total number of sampled configurations indicated above each distribution. In total, 4368 unique catalytic states were obtained from independent searches across different cluster sizes. The Boltzmann distribution of all the isomers within $\Omega = 0.4$ eV of the GM, representing the most accessible states for the neutral system, is shown in Figure 1b. Side and top views of the GM and low-lying LM structures included in our ensembles for further investigations are presented in Figure 1c and Figure S1. Note that the present ensemble-based screening approach selects nanocluster structures based on their thermodynamic proximity to the GM. While low-energy LMs are expected to be accessible under reaction conditions, interconversion between the GM and LMs may be kinetically constrained. In the presence of adsorbates, kinetic barriers may reshape the accessible isomer distribution under reaction conditions [14, 17, 18]. However, a rigorous evaluation of the kinetic barriers associated with GM $\rightarrow$ LM transitions, particularly in the presence of adsorbates, could further refine structure selection but would require extensive transition-state searches and dynamical simulations that lie beyond the scope of the present study. Nevertheless, our thermodynamic evaluation provides a robust first-principles basis for the data-driven screening of subnanometer electrocatalysts.

In the Pt_n/G ($n = 7 - 13$) size regime, subnano cluster shows diverse conformations with a distinct evolution in morphology, symmetry, and stability driven by the interplay of Pt–Pt bonding and Pt–C interactions on the graphene substrate. For Pt_7/G , GM adopts a compact, non-planar C_{3v} -symmetry structure [19], in contrast to the planar or prismatic motifs typically favored by unsupported or oxide-supported clusters [20, 21]. Similarly, the accessible isomers of Pt_8/G are dominated by bilayer and trilayer non-planar arrangements with shallow energy landscapes, reflecting their high fluxionality and dynamic behavior. The LM1 of Pt_8/G lies only 0.05 eV above the GM, underscoring its high accessibility within the ensemble-derived activity.



SCHEME 1 | Machine learning (ML) framework for catalyst discovery. The workflow consists of four streamlined steps: (1) global optimization of graphene-supported Pt_n ($n = 7 - 13$) subnano clusters to identify relevant global and local minima, followed by construction of an adsorption energy database for *O, *OH, and *OOH intermediates using DFT calculations; (2) feature space design to encode elemental, geometric, and electronic descriptors that capture the local coordination environment and structure-activity relationships in the subnanometer regime; (3) ML model evaluation and interpretability; and (4) uncertainty quantification, DFT validation, and ab initio phase diagrams under higher intermediate coverage and derivation of reaction networks.

For Pt_9/G , although planar geometries are preferred for unsupported Pt clusters due to relativistic effects [22], the supported GM shifts to a non-planar geometry, marking the onset of distinct 3D cluster character. As the size increases from Pt_{10}/G to Pt_{12}/G , the GM and LM adopt disordered tetrahedral and octahedral motifs, losing the “magic number” stability observed in vacuum as a result of strong d- π hybridization and binding with graphene [23]. For Pt_{13}/G , the GM becomes a distorted tricapped pentagonal prism or a distorted octahedron, in contrast to the icosahedral or high-symmetry structures typically observed for bare clusters. For further catalytic investigations, we constructed an ensemble of the most stable GM and the first five LMs, which are expected to have the highest population within $\Omega = 0.4$ eV of the GM, and performed a combined ML and DFT-based ORR mechanistic analysis on them.

3 | Results and Discussion

3.1 | Adsorption Energy Landscape of Pt_n/G Clusters

In line with the Bell-Evans-Polanyi (BEP) principle, the adsorption energy (E_{ads}) of the ORR intermediate serves as an important descriptor to evaluate the catalyst’s performance [24]. However, the inherent fluxionality of subnano clusters due to the presence of multiple heterogeneous active sites makes it challenging

to identify the most stable adsorption configurations on the local atomic environment. Therefore, to generate our initial E_{ads} database, we optimized multiple configurations of single-intermediate adsorbed on GM and LM isomers, while sampling different top and bridge positions for principal ORR intermediates (*O, *OH, and *OOH). The total number of configurations explored to construct the initial database is provided in Table S1. The E_{ads} for each intermediate was computed using the following Equation (1):

$$E_{ads} = E_{Pt_n/G}(*O, *OH, *OOH) - (E_{Pt_n/G} + E_{*O, *OH, *OOH}) \quad (1)$$

where $E_{Pt_n/G}(*O, *OH, *OOH)$, $E_{Pt_n/G}$, and $E_{*O, *OH, *OOH}$ represents the total energies of the cluster with *O, *OH, and *OOH, supported cluster and their corresponding intermediates, respectively. The E_{ads} trends for each intermediate are summarized in Figure 2a, illustrating the overall energy distribution sampled across various Pt_n/G sizes and heterogeneous sites. The E_{ads} values for each intermediate lack a consistent trend, with their peak energy distribution spanning distinct regions: -3.22 to -6.11 eV for *O, -1.97 to -4.64 eV for *OH, and -0.12 to -3.23 eV for *OOH. Compared to the bulk Pt(111) surface with $E_{*O} = -4.69$ eV, $E_{*OH} = -2.63$ eV, and $E_{*OOH} = -1.64$ eV [25], the subnano clusters exhibit a broad range of E_{ads} , with the majority showing stronger binding to the intermediate, attributed to the presence of under-coordinated active sites. Notably, the E_{ads} for Pt_8/G exhibits a

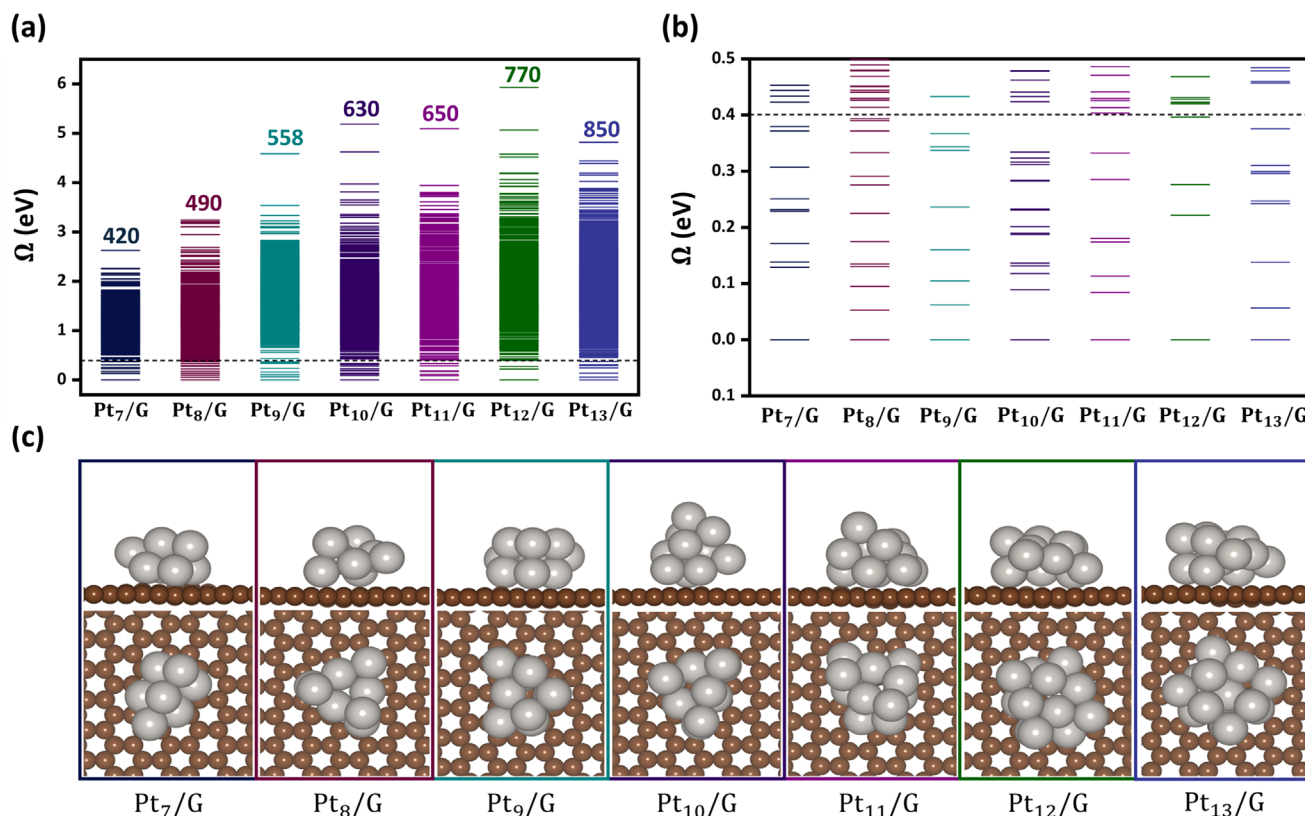


FIGURE 1 | Energy distribution from global optimization. (a) Isomeric energy distribution of the global minima (GM) and local minima (LMs) of Pt_n/G ($n = 7 - 13$) sampled in this study, with each state's energy referenced to that of the GM ($\Omega = E_{LM} - E_{GM}$, in eV). The number of configurations for each size regime is indicated above the corresponding distribution. The horizontal dashed line indicates the Boltzmann cutoff at $\Omega = 0.4$ eV. (b) Boltzmann distribution of all unique structures within 0.4 eV of the GM, representing the most accessible states for the neutral systems. (c) Side and top views of the most stable GM configurations for each cluster size, highlighting the size-dependent structural variations of these subnano clusters. The geometries of the LM states for each size regime, utilized to construct our ensemble representation within a $\Omega = 0.4$ eV difference for the ML-DFT analysis, are provided in Figure S1.

wide range of distribution for ORR intermediates across different isomers ($E_{*O} = -5.92$ eV for LM2; $E_{*OH} = -4.28$ eV for LM5; $E_{*OOH} = -2.82$ for LM2), compared to other Pt_n/G size regimes (Table S2). Figure 2b represents the energy distribution of the most stable single-intermediate configurations across different isomers in the ensemble ($E_{*O} = -5.92$ eV for LM2 of Pt_8/G , $E_{*OH} = -4.29$ eV for GM of Pt_{10}/G , and $E_{*OOH} = -2.84$ eV for LM1 of Pt_9/G) (Table S2). The side views of the optimized geometries for these configurations on the GM and LMs are shown in Figure 2c and Figure S2, respectively. In the subnanometer regime, the intermediate preferentially binds to the top and bridge sites, unlike on the Pt(111) surface, where the $*O$ intermediate favors adsorption at the hollow site [25]. Within this sampling space, we have curated a data set of 1275 data points with three intermediates (428 for E_{*O} , 423 for E_{*OH} , and 424 datapoints for E_{*OOH}), represented as “Dataset-1” in Figure 2d.

Owing to their multiple undercoordinated sites, every atom in subnano clusters exhibits a unique electronic structure and coordination geometry, resulting in significant variations in E_{ads} . According to the Sabatier principle [26], the intermediate should neither bind too strongly ($E_{*O} = -6.11$ eV as in Pt_{11}/G) leading to chemisorption, nor too weakly ($E_{*OOH} = -0.12$ eV as in Pt_{13}/G) leading to physisorption, to achieve high ORR activity (Table S2). After acquiring Dataset-1, we applied three screening

criteria based on the E_{ads} of each intermediate to screen the active electrocatalysts, as shown in Figure 2d. The upper and lower limits for E_{*O} , E_{*OH} , and E_{*OOH} were determined through a comprehensive analysis of various systems, including Pt_{1-10} subnano clusters, Pt_{79} nanocluster, and Pt(111) surface [6, 20, 25, 27, 28]. Additionally, to account for the metastability-triggered reactivity, weighted by the Boltzmann statistics, we extended each energy window by ± 0.4 eV (total extension of 0.8 eV), allowing us to preliminarily exclude the thermodynamically inaccessible states [6, 29–31]. Following this screening criterion, we curated Dataset-2, comprising 421, 422, and 416 datapoints for E_{*O} , E_{*OH} , and E_{*OOH} , respectively. Therefore, this E_{ads} dataset is uniformly sampled across catalysts with varying sizes, coordination environments, and three intermediates, which is pivotal for developing robust ML models.

3.2 | Feature Space Design for ML Models

Integrating both physical and chemical descriptors into ML models is critical for achieving high predictive accuracy. In this context, feature engineering plays an important role in model development, enabling the effective representation of complex local chemical environments while mapping the structure-activity relationships [9]. The 12 descriptors listed in Table 1 were

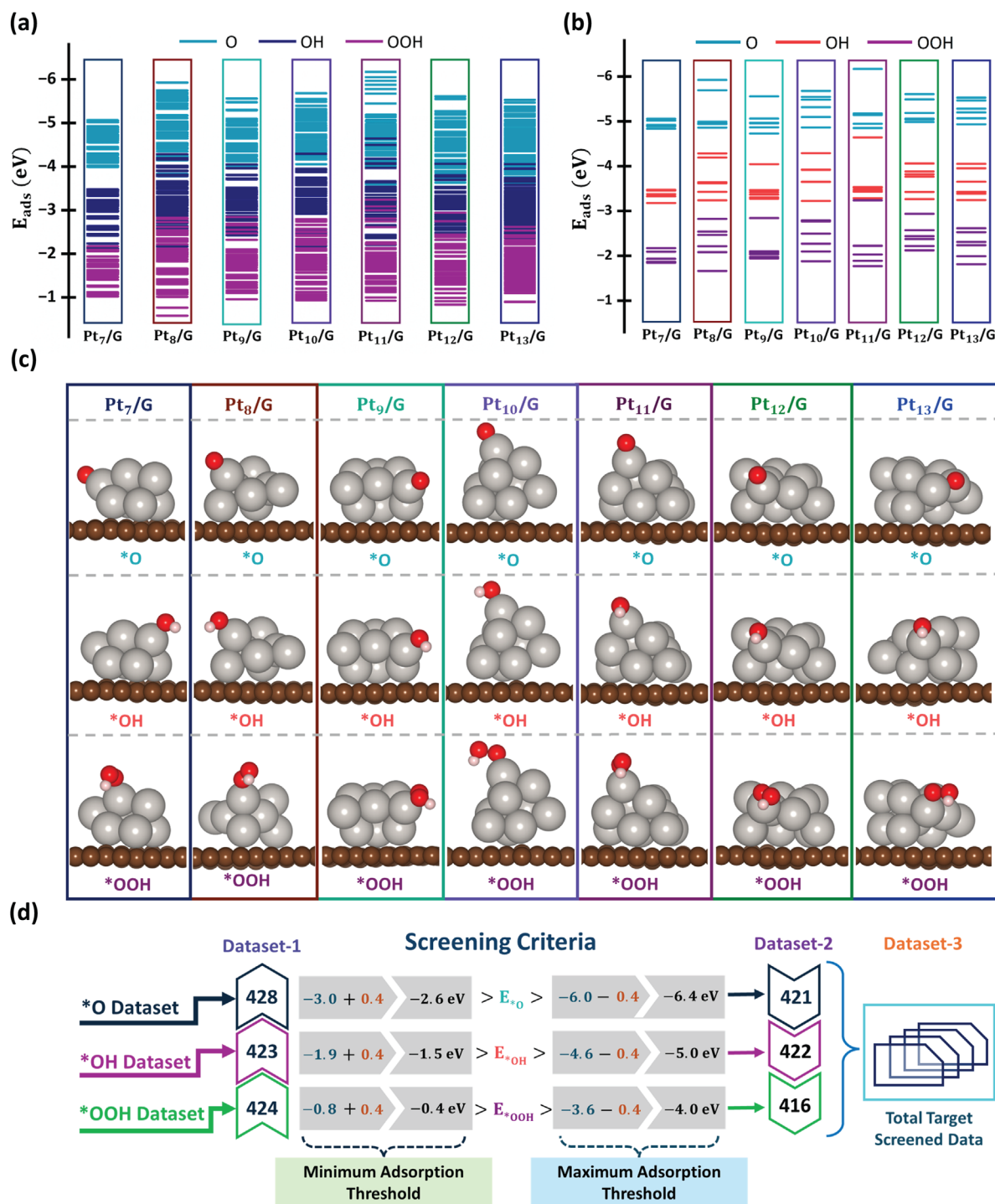


FIGURE 2 | Database generation for intermediate adsorption. (a) Adsorption energy distribution (E_{ads} , in eV) of key ORR intermediates (*O, *OH, and *OOH) on Pt_n/G subnano clusters ($n = 7-13$), sampled across global minima (GM) and low-lying local minima (LM) structures. The variations reflect sensitivity to cluster size, isomeric distribution, and coordination environment. (b) Distribution of E_{ads} for the most stable single-intermediate configurations across GM and LM isomers within 0.4 eV of the GM, based on Boltzmann statistics. (c) Side-view representations of the most favorable adsorption configurations for *O, *OH, and *OOH on GM structures of Pt_n/G clusters, where adsorbed species are denoted with an asterisk (*). Platinum, oxygen, and hydrogen atoms are shown in grey, red, and pink, respectively. Corresponding LM configurations are provided in Figure S2. (d) Schematic representation of the screening strategy used to extract optimal E_{ads} values for *O, *OH, and *OOH following the Sabatier principle, enabling identification of promising ORR electrocatalysts.

TABLE 1 | List of Primary Features including elemental, electronic, and geometric descriptors.

Groups	Descriptors	Symbol
Elemental Features	Number of Pt atoms	N
	Number of exposed Pt atoms of the cluster	M
Electronic Features	Sum of the number of d electrons	Σd_n
	The sum of the electronegativity of the cluster	$\Sigma \chi$
	The sum of the electron affinity of the cluster	ΣEA
	The sum of the atomic weights of the cluster	ΣAW
Geometric Features	Adsorption site (top, bridge)	S
	Coordination number	CN
	Perpendicular distance between adsorbate and surface	D_{AS}
	Perpendicular distance between the active Pt atom and the surface	Z
	Pt–O bond length	L
	Number of Pt atoms binding to the surface	A

selected to capture the factors governing intermediate adsorption on active sites of subnanometer clusters. We constructed a comprehensive set of input descriptors grouped into elemental, electronic, and geometric categories [9]. Elemental descriptors quantify cluster size and active-site density, electronic descriptors characterize the cluster's ability to donate or accept electron density during adsorption, and geometric descriptors define the local coordination environment. In this work, the term “surface” refers to the graphene support interacting with the Pt subnanoclusters, serving as the reference for defining adsorption geometry. In particular, surface-referenced geometric features, such as the perpendicular distances between the adsorbate and active Pt atoms (Z) and between the adsorbate and the support (D_{AS}) [32], capture metal–adsorbate and metal–support interactions. Such combined geometric-electronic feature sets are widely encoded in the ML models to uncover the structure-activity relationships. Although DFT calculations remain indispensable for generating accurate geometric and electronic descriptors, ML plays a crucial role in optimizing, interpreting, and exploring the resulting descriptor space. ML enables the systematic identification of sparse and non-redundant descriptor subsets that retain high predictive accuracy while remaining physically interpretable [9, 33]. This ML-driven descriptor analysis allows us to isolate the specific descriptors responsible for the breaking of conventional scaling relations in subnanometer Pt clusters, thereby revealing the structural origins of non-scaling behavior that cannot be systematically extracted from direct DFT analysis alone. Beyond interpretability, ML also enhances computational efficiency by predicting adsorption energetics and ORR activity across large configurational spaces, enabling rapid screening of the most active electrocatalysts, which are subsequently validated through targeted DFT calculations.

Further, to derive a rational and non-redundant feature set for ML investigation, we evaluated both linear and monotonic relationships between input features and the target E_{ads} using Pearson's correlation coefficient (PCC) and Spearman's rank correlation coefficient (SCC) [34, 35], as shown in Figures S3–S6 and detailed in Texts S2 and S3. To minimise redundancy, we retained only those features with low feature-feature correlation

($|PCC| < 0.8$ and $|SCC| < 0.8$) and high correlation with the target variable. The final set of selected descriptors is provided in Table S3. The target E_{ads} for each intermediate can be expressed as a function of the most relevant features:

$$E_{Pt_n/G}^{*O, *OH, *OOH} = f(\Sigma d_n, S, CN, D_{AS}, Z, L, A) \quad (2)$$

3.3 | Model Training and Performance Evaluation

Following feature engineering, model training was carried out to optimise the parameters of various ML algorithms to minimise the discrepancy between predicted and actual E_{ads} . Dataset-2 was split into training and testing subsets using an 80:20 ratio. Hyperparameter tuning was performed using both grid search and randomised search strategies, each integrated with 5-fold cross-validation to identify optimal models. Model performance was evaluated based on the lowest mean absolute error (MAE), root mean square error (RMSE), and the highest coefficient of determination (R^2) values (Text S5) [36]. To ensure robust and generalizable prediction across the chemically diverse subnano cluster space, seven regression algorithms were employed: kernel ridge regression (KRR), eXtreme gradient boosting regression (XGBR), random forest regression (RFR), adaptive boosting regression (ABR), extra trees regression (ETR), gradient boosting regression (GBR), and CatBoost regression (CR). All models were implemented using the open-source Scikit library (Text S5). Given the complexity and heterogeneity of the subnanometer catalyst, the combination of kernel-based and ensemble-based methods offers a balanced strategy capable of capturing both linear and non-linear relationships in the data. Model performance metrics are summarized in Figure 3, and the corresponding hyperparameters are detailed in Tables S4–S6.

In Figure 3, panels (a–c) show the comparative test performance of various ML models for predicting the E_{ads} of the three key intermediates. Due to variations in the statistical distribution and feature-target relationships across the datasets, different models performed best for different intermediates. The GBR model showed the highest accuracy in predicting E_{*O} and E_{*OOH} with

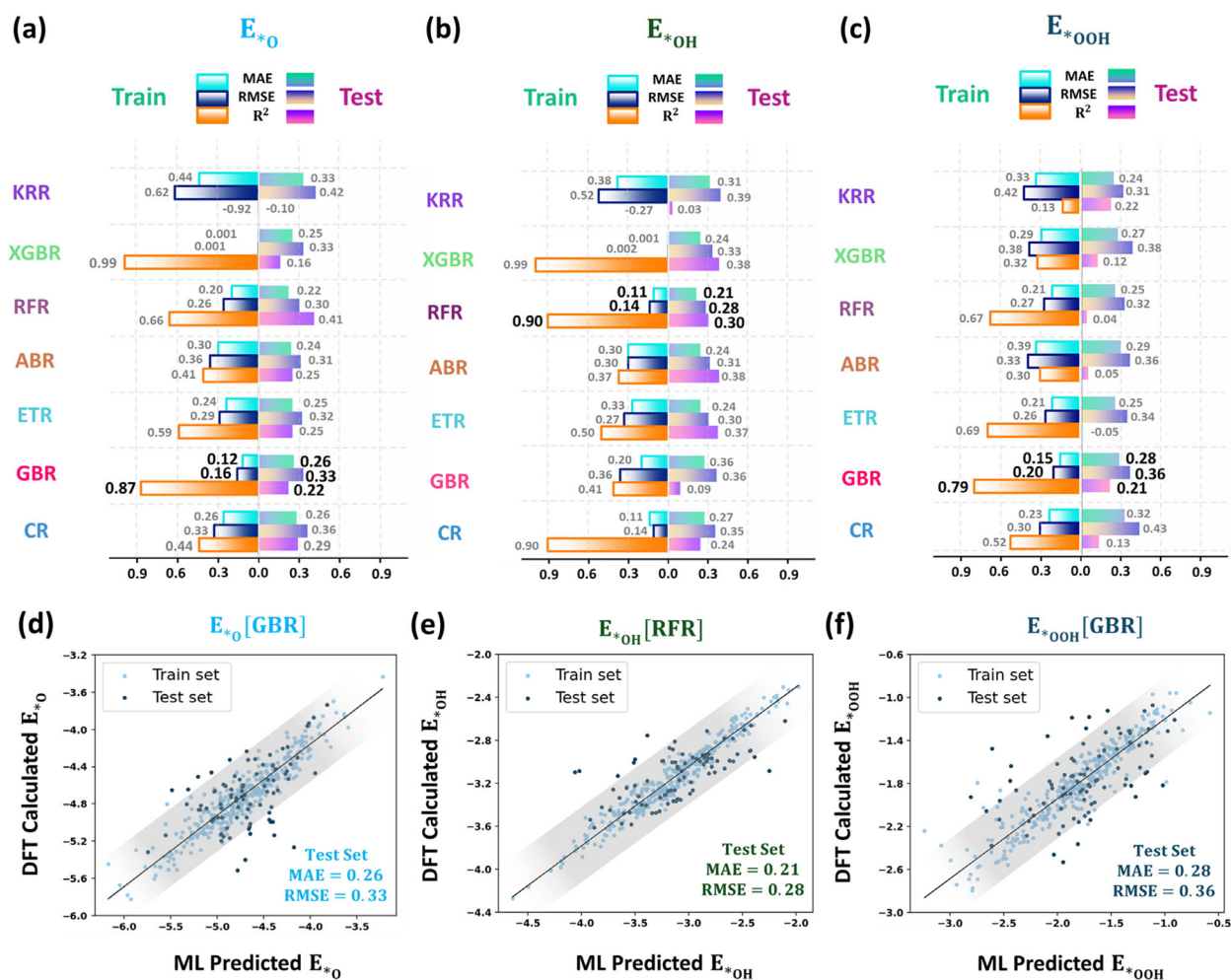


FIGURE 3 | ML model training and evaluation. Mean absolute error (MAE) and root-mean-square error (RMSE) for KRR, XGBR, RFR, ABR, ETR, GBR, and CR models using (a) E_{*O} , (b) E_{*OH} , and (c) E_{*OOH} datasets (eV units). The best-performing model for each intermediate is highlighted in bold. Parity plots comparing DFT-calculated and ML-predicted values for (d) E_{*O} , (e) E_{*OH} , and (f) E_{*OOH} , obtained from the respective best-performing models.

MAE/RMSE values of 0.26/0.33 and 0.28/0.36 eV, respectively, for the test datasets. In contrast, the RFR outperformed other models for E_{*OH} with an MAE/RMSE of 0.21/0.28 eV for the test dataset. Notably, we have achieved this accuracy for E_{*O} , E_{*OH} , and E_{*OOH} values across different heterogeneous sites of Pt_n/G subnano clusters ($n = 7-13$). The predictive reliability of these models is further supported by the parity plots in panels (d–f) of Figure 3, which display correlations between DFT-calculated and ML-predicted E_{ads} for both training and testing sets. Based on the above ML model performances, GBR was selected for predicting E_{*O} and E_{*OOH} , while RFR was selected for predicting E_{*OH} in the subsequent E_{ads} evaluations and ORR mechanistic pathway analysis. Although a single ML model could, in principle, predict adsorption energies of multiple ORR intermediates, this typically requires large and highly diverse datasets [37, 38]. Under the present data regime, adsorbate-specific models perform more reliably due to the distinct bonding characteristics and electronic interactions of different intermediates, which lead to varying degrees of non-linearity and noise in the structure-property relationships. Consequently, a universal model trained on mixed adsorbate data must balance competing trends, resulting in reduced predictive accuracy compared to dedicated models.

3.4 | Feature-Driven Interpretability of Low Scaling Relationships

Note that subnano clusters can readily adapt their shapes and binding sites in response to different adsorbates, which gives rise to non-Arrhenius behavior [6]. To examine this, we analyzed the correlations between E_{*O} vs. E_{*OH} , and E_{*OH} vs. E_{*OOH} for both ML (Figure 4a,b) and DFT (Figure S7) datasets. The consistently low R^2 values across different size regimes indicate the breaking of scaling relationships in the subnanometer regime. This non-scaling behavior arises from the high fluxionality of the clusters and the presence of multiple heterogeneous active sites on their irregular geometries, in contrast to bulk surfaces, where $R^2 = 0.91$ [8].

To identify the factors responsible for this deviation, we performed a feature-driven uncertainty analysis to identify the size regimes and key geometric descriptors. Figure 4c shows that the E_{*O} , E_{*OH} , and E_{*OOH} are spread across distinct and irregular regions, further reflecting this heterogeneity. We then carried out a leave-one-feature-out analysis on Pt_n/G , where each size regime was sequentially excluded, and the model performance

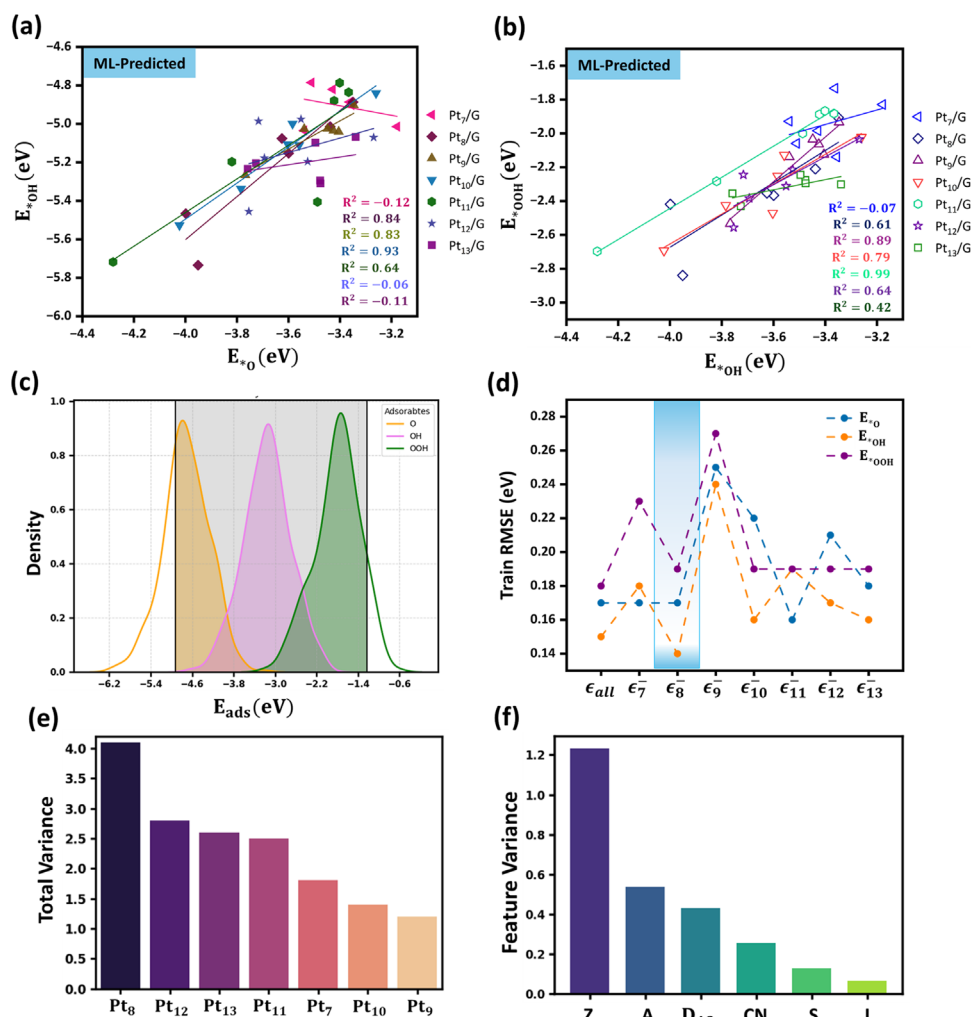


FIGURE 4 | Scaling Relationships and quantitative evaluation of ML model uncertainty. (a) ML-predicted scaling relationship between (a) E_{*O} vs. E_{*OH} and (b) E_{*OH} vs. E_{*OOH} for Pt_n/G ($n = 7 - 13$) subnano clusters. The coefficient of determination (R^2) of the best-fitted line is displayed on the right side of each plot. The DFT-predicted scaling relationship between E_{*O} vs. E_{*OH} and E_{*OH} vs. E_{*OOH} is represented in Figure S7. (c) Density distribution of E_{ads} for ORR intermediates. (d) Training RMSE values of the GBR model upon systematic exclusion of individual cluster sizes. The shaded bar highlights the minimum error observed for different intermediates when Pt₈/G clusters are excluded from Dataset-2 (denoted as ϵ_8^-). Additional metrics, including test RMSE, train MAE, and test MAE for the sequential removal of each size regime, are provided in Figure S8. (e) Bar plots showing the variance contributions in the GBR model across cluster sizes, based on descriptor-driven cluster behavior within the optimal E_{ads} window ($-4.92 < E_{ads} < -1.23$ eV). (f) Quantification of geometric feature-based variance in the Pt₈/G isomers dataset.

(RMSE and MAE) was re-evaluated for E_{*O} , E_{*OH} , and E_{*OOH} (Figure 4d; Figure S8). Notably, excluding Pt₈/G clusters (denoted as ϵ_8^-) consistently lowered both training and test errors, whereas removing other sizes had a minimal effect. Unlike bare Pt₈ clusters with planar geometries [19], Pt₈/G isomers are especially heterogeneous, contributing most to the low R^2 values. Their broad and inconsistent descriptor distribution acts as noise, reducing the overall model's robustness.

We further analyzed the combined variance of Pt_n/G clusters within an optimal E_{ads} window ($-4.92 < E_{ads} < -1.23$ eV) relevant for ORR intermediates (Figure 4e). Again, Pt₈/G displayed the highest variance, strongly linked to overfitting and reduced predictive accuracy compared to other cluster sizes. To interpret the underlying cause of this uncertainty, we performed a feature-level variance analysis of Pt₈/G systems (Figure 4f). Geometric descriptors, specifically Z, A, and D_{AS}, contribute

the most to the variance, followed by CN (Table 1). Among these, Z emerges as the single most influential descriptor, reflecting its dual role in encoding cluster height (anchoring of graphene) and electronic coupling with the substrate. This explains the heterogeneity observed in Pt₈ clusters anchoring, spatial Pt atom arrangements, and coordination environments across GM and LM isomers, which collectively degrade model performance.

Although ML models capture complex non-linear patterns effectively, they often lack interpretability, reinforcing their “black-box” perception. To improve transparency, we applied TreeExplainer-based SHapley Additive exPlanations (SHAP) analysis [39], a cooperative game theory approach that assigns a contribution value to each feature for a given prediction. SHAP analysis provides both global and local interpretability, offering insights into feature importance and model decision pathways for

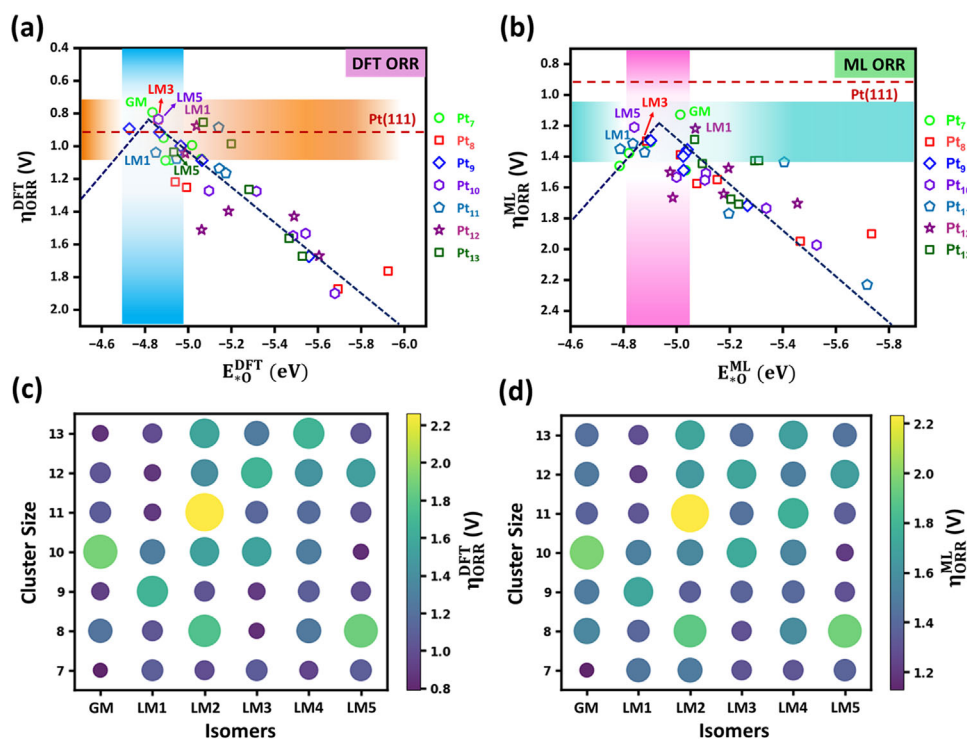
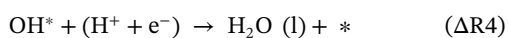
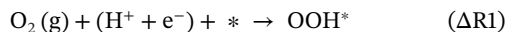


FIGURE 5 | ORR volcano plot and activity mapping. Volcano plots showing the ORR performance of subnano clusters based on (a) DFT-calculated overpotentials ($\eta_{\text{ORR}}^{\text{DFT}}$) and (b) ML-predicted overpotential ($\eta_{\text{ORR}}^{\text{ML}}$) for electrocatalysts in Dataset 3. Catalysts near the apex, located at the intersection of the dual-gradient bands, exhibit the lowest $\eta_{\text{ORR}}^{\text{DFT}}$ and $\eta_{\text{ORR}}^{\text{ML}}$ values, indicating high ORR activity. The horizontal dashed line represents the overpotential of the Pt(111) surface ($\eta = 0.92$ V), taken from our previous investigation for comparison [25]. Panels (c) and (d) show bubble plots of DFT-calculated $\eta_{\text{ORR}}^{\text{DFT}}$ and ML-predicted $\eta_{\text{ORR}}^{\text{ML}}$ values, respectively, across different cluster sizes and isomers.

E_{ads} . A detailed SHAP-based global feature attribution analysis is provided in Text S6 and Figure S9.

3.5 | Validation of ML-Predicted Adsorption Energy

To explore the catalytic activity of all the isomers extracted in our ensemble, we considered the most stable intermediate adsorption configuration from the DFT-calculated and their corresponding ML-predicted values from the E_{ads} dataset. Subsequently, we derive the ORR mechanism via an associative pathway to identify the active electrocatalysts, compiled as Dataset-3 in our investigation (Figure 2d). We utilized the computational hydrogen electrode (CHE) model proposed by Nørskov and co-workers to calculate the reaction energy at 0 and 1.23 V, following the elementary steps outlined below [40]:



The ORR activity of the subnano clusters in the ensemble is evaluated by the overpotential values (η) for the rate-determining step (RDS) at 1.23 V, as listed in Table S7, using the following equations:

$$\eta_{\text{ORR}}^{\text{DFT/ML}} = \frac{\max(\Delta\text{R}_1, \Delta\text{R}_2, \Delta\text{R}_3, \Delta\text{R}_4)}{e} \quad (3)$$

The variable distribution of the DFT-calculated $\eta_{\text{ORR}}^{\text{DFT}}$ and ML-predicted $\eta_{\text{ORR}}^{\text{ML}}$ values for the RDS (Figure S10) underscores the non-monotonic ORR activity in the subnanometer regime with size and isomers. Specifically, the GM of Pt₇/G, LM3 of Pt₈/G, and LM5 of Pt₁₀/G emerged as the active electrocatalysts with lower $\eta_{\text{ORR}}^{\text{DFT}}$ values of 0.79, 0.83, and 0.83 V, respectively, with RDS involving $\text{*OH} \rightarrow \text{H}_2\text{O}$ formation (ΔR_4) for each catalysts (Table S7). In both the DFT and ML datasets (Figure 5a,b), these catalysts lie near the apex of the volcano plots. Figure 5c,d presents bubble plots highlighting the similarity in the trends of $\eta_{\text{ORR}}^{\text{DFT}}$ and $\eta_{\text{ORR}}^{\text{ML}}$ values for all catalysts in our ensemble across each size range at 1.23 V, while Figure S11, shows the reaction energy diagrams for all catalysts at 0 and 1.23 V. Compared to bare Pt₇ and Pt₁₃ clusters, the supported Pt₇/G and Pt₁₃/G isomers exhibits higher η values in both DFT and ML dataset [20, 30].

For enhanced ORR performance in fuel cells, the catalyst at the apex of the activity volcano should exhibit near-optimal E_{ads} for each intermediate [41]. Upon closer examination, the DFT-calculated and ML-predicted $E_{\text{*O}}$, $E_{\text{*OH}}$, and $E_{\text{*OOH}}$ values of the best electrocatalysts at the volcano peak fall within an optimal

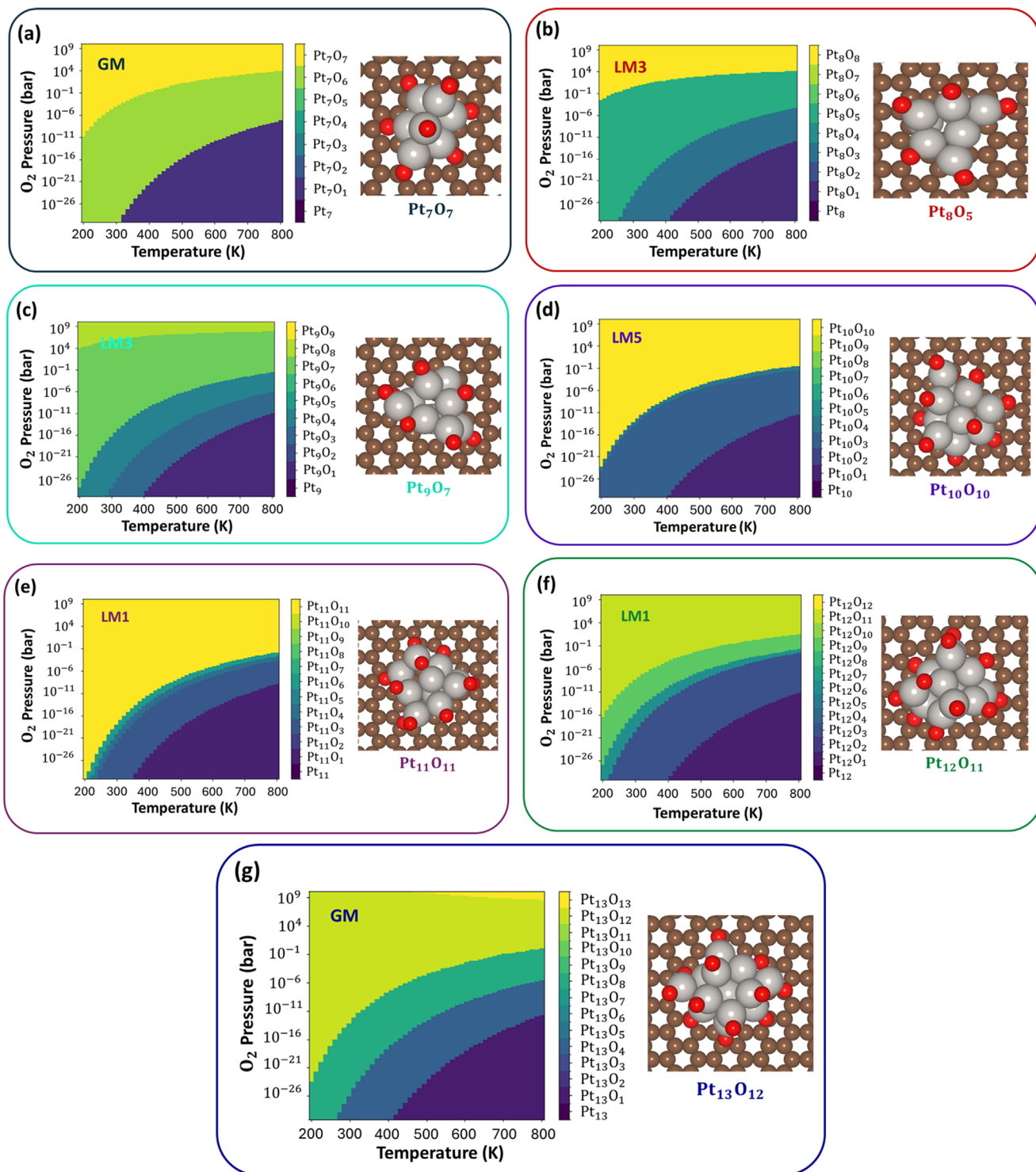


FIGURE 6 | Ab initio thermodynamic analysis. Phase diagrams for active isomers illustrating the evolution of stable *O coverage configurations. Panels (a–g) correspond to Pt_7O_x ($x = 1-7$), Pt_8O_x ($x = 1-8$), Pt_9O_x ($x = 1-9$), Pt_{10}O_x ($x = 1-10$), Pt_{11}O_x ($x = 1-11$), Pt_{12}O_x ($x = 1-12$), and Pt_{13}O_x ($x = 1-13$). For each cluster size, the most stable high-coverage geometries under ambient ORR conditions are shown alongside the corresponding phase diagram. The remaining configurations of *O coverage are shown in Figure S12.

energy range, closely matching the predefined E_{ads} interval defined by the equation:

$$-4.92 \text{ eV} < E_{\text{ads}} < -1.82 \text{ eV} \quad (4)$$

Within this specified range, the active subnano clusters exhibit

optimal binding affinities with each intermediate and emerge as the most promising candidates (Table S8). This E_{ads} range in the subnanometer regime can therefore be used to directly identify active electrocatalysts, potentially overcoming the conventional scaling relationship for ORR. Note that the Pt cluster sizes investigated in this work are not intrinsically limited,

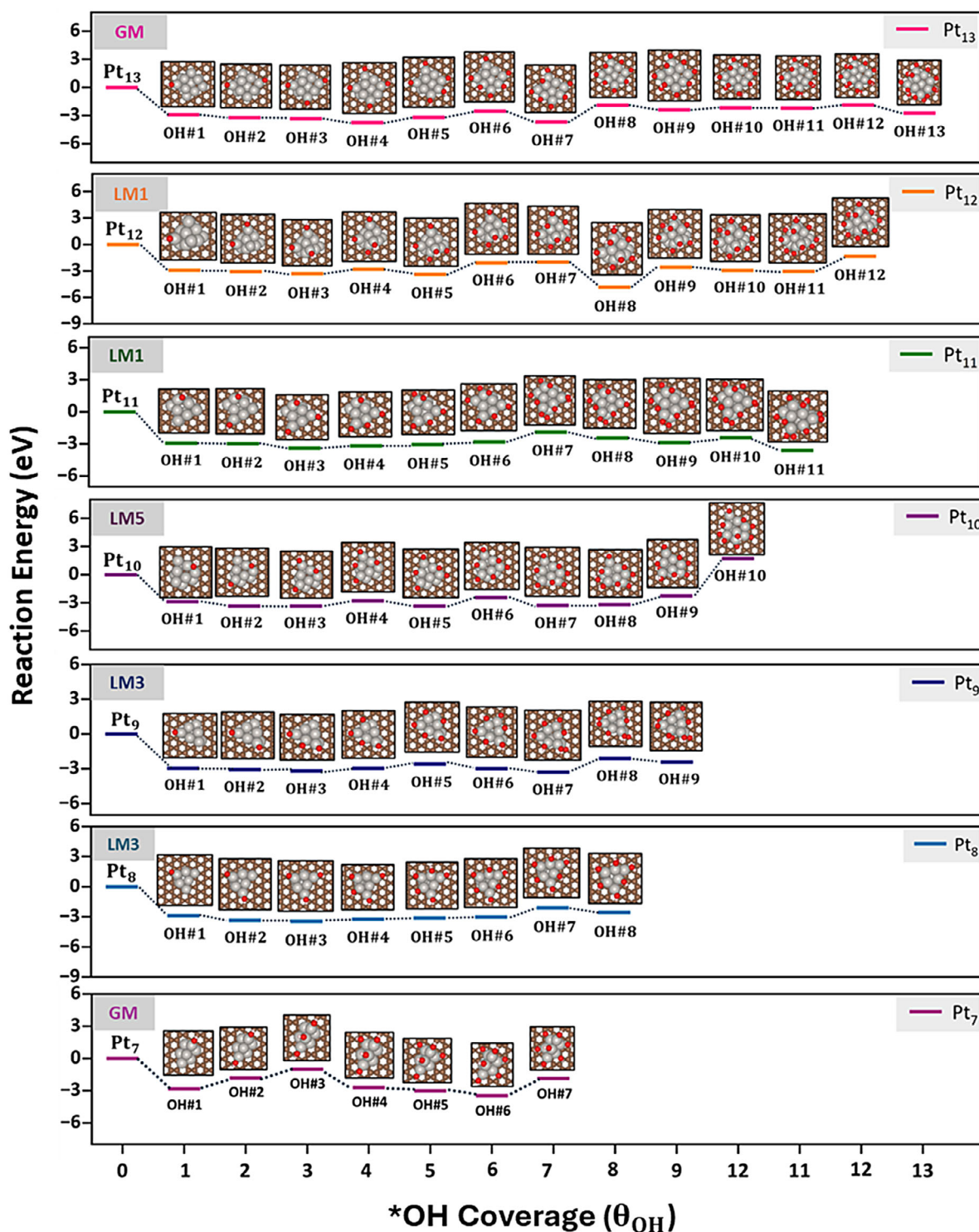


FIGURE 7 | Reaction energy diagram at higher *OH coverage (θ_{OH}). Schematics of the hydroxylation process on the active Pt_n/G cluster catalyst leading to complete monolayer *OH coverage: Pt₇OH_x (x = 1–7), Pt₈OH_x (x = 1–8), Pt₉OH_x (x = 1–9), Pt₁₀OH_x (x = 1–10), Pt₁₁OH_x (x = 1–11), Pt₁₂OH_x (x = 1–12), and Pt₁₃OH_x (x = 1–13). For each cluster size, the most stable high-coverage geometries and their reaction networks (e.g., Pt_nOH_x and Pt_nOH_{x+1}) are shown alongside the corresponding reaction energy diagram.

and the proposed framework can be systematically extended to graphene-supported Pt nanoclusters with different sizes and compositions. Consistent with our earlier studies on bare Pt₁₃ clusters [30] and graphene-supported Pt_{7,8} clusters [28], the same global optimization strategy, descriptor construction, and ML

workflow can be applied to additional Pt clusters to predict their ORR performance. Such an extension would enable a systematic assessment of size-dependent ORR activity trends and provide a future direction for exploring larger cluster sizes and more realistic reaction conditions.

3.6 | Ab Initio Thermodynamic Analysis

Since multiple cluster configurations are expected to coexist under reaction conditions, we investigated the nature of active isomers at higher intermediate coverage. Under experimental ORR conditions, Pt clusters are prone to oxidation, forming oxide-driven structures at ambient temperatures and pressures relevant to catalysis [42]. To capture these effects, we developed an ab initio thermodynamic framework to analyze the oxidation thermodynamics of Pt_nO_x ($n = 7-13$, $x = 1-13$). For each size and coverage, we sample the configurational space by generating oxide-based structures, followed by DFT optimization. In this framework, the chemical potential of gas-phase oxygen was treated as a function of temperature and pressure, $\mu_{\text{O}}(T, P)$, expressed as:

$$\mu_{\text{O}}(T, P) = \frac{1}{2} \left[E_{\text{O}_2}(T = 0\text{K}, P^0) + \mu_{\text{O}_2}(T, P^0) + k_B \ln \left(\frac{P}{P^0} \right) \right] \quad (5)$$

where T is temperature, P is the partial pressure of oxygen, P^0 is the standard atmospheric pressure, and k_B is the Boltzmann constant. Vibrational contributions from Pt and O atoms were included in the free energy calculations. The Python Multi-scale Thermochemistry Toolbox (pMuTT) was used to compute temperature- and pressure-dependent thermodynamic corrections from vibrational frequency analysis [43]. The resulting phase diagram and the stable configurations at ambient ORR conditions are shown in Figure 6, and the remaining configurations of *O coverage are shown in Figure S12.

All isomers undergo significant geometric transformations at the core under high *O coverage, indicating fluxionality under reaction conditions. Unlike the recent report by Zhang et al., the Pt_nO_x clusters in each size regime remain intact [44], without oxygen insertion into the core or sintering of Pt-O motifs, highlighting their stability. At higher coverage, O intermediates predominantly occupy top sites in each configuration, in contrast to the more diverse adsorption sites observed at lower coverage. Under ambient ORR conditions ($T = 300\text{ K}$, $P_{\text{O}_2} = 1\text{ bar}$), the isomers tend to form a complete oxygen monolayer, suggesting a strong oxidation tendency. The extent of this coverage may vary with experimental parameters such as electrolyte concentration and pH, potentially leading to even more pronounced structural changes in the subnano clusters. Overall, our results demonstrate that increasing oxygen coverage, coupled with cluster restructuring, drives adsorbate-dependent fluxionality across different sizes and provides a systematic framework to model these complexities.

3.7 | Ensemble-Level Understanding of Reaction Networks

Since the RDS involves $^*\text{OH} \rightarrow \text{H}_2\text{O}$ formation (ΔR_4) at low intermediate coverage for different isomers, it is important to derive their ORR performance at higher *OH coverage. To account for this, we extend our analysis to include the progressive addition of *OH intermediates to the cluster ensemble. The configurations for higher *OH coverages were derived using Yan et al.'s bond model [45], which allows construction of a reaction network by connecting neighboring structures with successive

*OH coverage (e.g., Pt_nOH_x and $\text{Pt}_n\text{OH}_{x+1}$), as illustrated in Figure 7 [46]. The pairwise energetics were then calculated, and the reaction energies for each elementary step were plotted, assuming full monolayer formation. Within this framework, the reaction barriers are assumed to scale linearly with the reaction energies, following the BEP relationship.

Note that for each size regime, adsorption of the first OH#1 is thermodynamically favorable, revealing that the real catalytic interface will be initially covered by *OH. Unlike the irregular reaction energies seen for successive *OH additions on Pt_7/G , the $\text{Pt}_{8-11}/\text{G}$ isomers show more stable reaction energies, with the most endothermic steps at OH#6 $\rightarrow$ OH#7, OH#7 $\rightarrow$ OH#8, OH#9 $\rightarrow$ OH#10, and OH#6 $\rightarrow$ OH#7, respectively. For $\text{Pt}_{10-13}/\text{G}$, the endothermic step begins with the formation of OH#4, OH#4, OH#5, and #OH5, marking the inflexion point corresponding to the onset of saturation, which is often accompanied by structure rearrangements in the clusters and a shift in adsorption from top to bridge sites. Beyond this point, the further *OH addition results in steric hindrance and lateral interactions among neighboring adsorbates, which further hinder stabilization. Consequently, the ORR activity diminishes at high *OH coverages, underscoring the importance of adsorbate-induced fluxionality and coverage effects in determining the catalytic performance of subnano clusters.

4 | Conclusions

In summary, our combined DFT-ML investigations of Pt_n/G ($n = 7-13$) identify the geometric factors of supported nanoclusters that govern the breaking of scaling relationships in ORR. The perpendicular distance between the active Pt atom and the graphene surface emerges as the most influential descriptor underlying the observed non-Arrhenius behavior. Furthermore, feature interpretability reveals that Pt_8/G isomers exhibit the highest structural heterogeneity, which introduces noise and reduces model robustness. Using a data-driven framework, we capture the trends in ORR activity from both ML-predicted and DFT-calculated overpotentials within the catalyst ensemble. The trained ML models, based on accessible non-ab initio geometric descriptors, achieve reasonable accuracy in predicting *OOH, *O, and *OH adsorption energies (MAE = 0.28, 0.26, and 0.21 eV, respectively). Thermodynamic analysis indicates that graphene-supported clusters favor complete *O monolayer formation, leading to relatively high overpotentials compared to bare clusters. Overall, this study highlights the structural origins of non-monotonic ORR activity in the subnanometer regime for their fuel cell applications.

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Conflicts of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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